MILITARY SPECIFICATION

AIR, LIQUID, FOR BREATHING PURPOSES

1. SCOPE

1.1 This specification covers liquid air of two grades to be used primarily for breathing purposes and which is produced by either direct liquefaction of air or synthesized from the mixing of liquid oxygen and liquid nitrogen. See intended purpose for more detailed data (6.1).

1.2 Classification. Liquid air shall be of the following grades as specified (6.2).

Grade A - High purity

Grade B - Ordinary purity

2. APPLICABLE DOCUMENTS

2.1 Defense standardization documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

BB-N-411	Nitrogen, Technical.
BB-O-925	Oxygen, Technical, Gas and Liquid.

Military

MIL-P-25508	Propellant, Oxygen.
MIL-O-27210	Oxygen, Aviators' Breathing, Liquid and Gas.
MIL-P-27401	Propellant, Pressurizing Agent, Nitrogen.
MIL-C-45662	Calibration Systems Requirements.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.

FSC 6830

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MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-172	Color for Containers of Liquid Propellants.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Department of Transportation (DOT).

49 CFR 170	Administrative procedures of the Department of Transportation.
49 CFR 171-190, 46 CFR 146, & 14 CFR 103.	Hazardous Materials Regulations of the Department of Transportation. (For use of alternate commercial publications see 6.5).

Department of Defense (DOD).

DSAM 4145.3; Also AFM 71-4, TM 38-250, NAVAIR 15-03-500, or MCO 4145.3.	Packaging and Handling of Dangerous Materials for Transportation by Military Aircraft.

National Bureau of Standards (NBS).

NBS Research Paper No. 1865	Measurement of Water in Gases by Electrical Conduction in a Film of Hygroscopic Material and the Use of Pressure Changes in Calibration.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

American Society for Testing and Materials (ASTM).

D2504-67	Test for Noncondensable Gases in C_3 and Lighter Hydrocarbons by Gas Chromotography.
D2543-66T	Sampling Cryogenic Aerospace Propellants.

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E29-67	Recommended Practice for Indicating Which Places of Figures Are to be Considered Significant in Specified Limiting Values.
E168-67	Recommended Practice for General Techniques of Infrared Quantitative Analysis.
E260-65T	Recommended Practice for General Gas Chromotography Procedures.
E334-68	Recommended Practice for General Techniques of Infrared Microanalysis.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3. REQUIREMENTS

3.1 Production. The material furnished under this specification shall be produced from either direct liquefaction of air or synthesis by mixing of liquid oxygen and liquid nitrogen. The starting constituents shall conform to BB-N-411, BB-O-925, MIL-P-25508, MIL-O-27210, MIL-P-27401, or any other commercially available liquid oxygen and liquid nitrogen; however, the finished product shall meet the requirements of this specification.

3.2 Properties. The liquid air shall conform to the requirements of table I when tested in accordance with the applicable test methods.

3.3 Limiting values. The following applies to all specified limits in this specification. For purposes of determining conformance with these requirements, an observed value or a calculated value shall be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limitation value, in accordance with the rounding-off method of ASTM Recommended Practice E29-67.

3.4 Filling.

3.4.1 Container processing. Containers shall be prepared, cleaned, tested, repaired, and filled in accordance with section 5 herein.

3.4.2 Filter. A filter with a 10-micron nominal and 40-micron absolute rating shall be installed between the outlet of the contractor's plant system and the inlet of the container to be filled for delivery. The element of the filter shall be so constructed of material which cannot flake-off or spall-off and migrate down stream to contaminate the containers to be filled.

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Table I Properties Limits

Component		Grade A (high purity)	Grade B (ordinary purity)	Test Methods
Assay				
Oxygen (O_2) - percent by volume.		19.5-23	19.5-23	4.6.1
Nitrogen (N_2) - percent by volume.		balance	balance	- - - -
Impurities				
Odor		none	none	4.6.2
Gaseous hydrocarbons (methane, ethane, etc.) - ppm by volume maximum as methane equivalent.		25	60 (*)	4.6.3
Unsaturated Hydrocarbons	Acetylene - ppm by volume maximum.	0.1	- - - -	4.6.4
	Ethylene - ppm by volume maximum.	0.4		
Halogenated hydrocarbons (trichloroethylene, etc.) - ppm by volume maximum.		1	- - - -	4.6.5
Carbon monoxide (CO) - ppm by volume maximum.		5	- - - -	4.6.6
Carbon dioxide (CO_2) - ppm by volume maximum.		5	- - - -	4.6.7
Nitrogen dioxide (NO_2) - ppm by volume maximum.		0.5	- - - -	4.6.8
Sulfur dioxide (SO_2) - ppm by volume maximum.		0.5	- - - -	4.6.9
Moisture - dew point ^{o}F maximum.		-65	- - - -	4.6.10
Particulate - mg/liter liquid maximum.		1.0	- - - -	4.6.11

NOTE:
(*) Test not required when product is produced exclusively from MIL-P-27401 and MIL-P-25508 or MIL-O-27210.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the product shall be classified as quality conformance tests.

4.3 Inspection of the product.

4.3.1 Lotting. A lot shall consist of one of the following:

(a) The product produced in not more than 24 consecutive hours from a continuous process which is used to fill shipping containers directly from the process output. A continuous process shall be the production of product by continuous input of raw materials and output of finished product by one manufacturer in one plant with no change in manufacturing conditions or materials.

(b) The product from individual runs of a batch process which is used to fill shipping containers directly from the process output. A batch process shall be the production of product by runs from single additions of raw materials which are mixed, reacted, and purified as necessary forming the product.

(c) The product from either or both the continuous and batch processes which is held in a single storage tank and subsequently withdrawn to fill shipping containers. The product shall be homogeneous at the time of withdrawal and shall not be added to while being withdrawn. After each addition to the storage tank the contents shall constitute a separate lot.

4.3.2 Sampling for test. Unless otherwise specified by the procuring activity, samples shall be withdrawn and tested at the point of manufacture and filling as there is no suitable sampler for shipping small quantities of air in the liquid phase which is required for certain of the quality conformance tests. Unless otherwise specified, sampling shall be performed in accordance with ASTM D2543-66T. Samples shall be taken directly from the shipping containers, except that with prior approval of the procuring activity the samples may be taken from the proximity of the contractor's filling outlet. For suitable samplers see 6.3. Samples shall be withdrawn within 24 hours of time for delivery to destination.

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4.3.3 Sampling plan.

4.3.3.1 Cylinders. The number of cylinders (dewar type) selected for sampling from each lot shall be in accordance with table II. The first and last cylinders to be filled within a given lot shall be sampled. Other samples shall be selected at random. The product from each cylinder sampled shall constitute a separate sample.

Table II Sampling Plan for Test

Number of cylinders in lot	Number of cylinders to be sampled
1-9	1
10-49	2
50-149	3
150 and over	4

4.3.3.2 Bulk. Each bulk transport (i.e., portable tanks, cargo tanks, and tank cars) shall be sampled at least once.

4.3.4 Rejection. When any sample of the product tested as specified herein fails to conform to the requirements specified herein, the entire lot represented by the sample shall be rejected.

4.4 Inspection of preparation for delivery.

4.4.1 Lotting. A lot shall consist of the total number of cylinders prepared for delivery for one order at one time, or each bulk transport.

4.4.2 Sampling plan. Sampling for examination of preparation for delivery shall be conducted in accordance with MIL-STD-105 to inspection level S-2 with an AQL of 4.0 percent defective. The sample unit shall be one filled unit, intermediate, or shipping container as applicable.

4.4.3 Examination. Samples selected in accordance with 4.4.2 shall be examined for compliance with the requirements of section 5 herein and any additional packaging requirements specified in the contract or purchase order. In addition, there shall be evidence of the acceptance of the product lot(s) represented; otherwise, the lot(s) offered for delivery shall be rejected.

4.5 Test conditions. The test conditions herein specified shall apply to the test methods in 4.6.

4.5.1 Chemicals. Unless otherwise specified, all chemical tests shall be made with A.C.S. specification reagent grade chemicals and distilled water.

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4.5.2 Calibration gases. Calibration gas standards required for calibration of analyzers or other test apparatus shall be of certified purity and analysis.

4.5.3 Calibration. Test equipment shall be calibrated as applicable in accordance with MIL-C-45662.

4.5.4 Operations. Unless otherwise specified, analytical equipment shall be operated in accordance with the manufacturer's instructions.

4.6 Test methods.

4.6.1 Oxygen assay. The oxygen assay shall be determined in percent by volume by one of the following methods. In cases of dispute the method in (b) shall be referee.

(a) By a paramagnetic type analyzer with an accuracy at least 0.5 percent. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals by the use of calibration gas standards using nitrogen as the base gas.

(b) By an electrochemical type analyzer with an accuracy at least 0.5 percent and containing a solid or aqueous electrolite. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals by the use of calibration gas standards using nitrogen as the base gas.

4.6.2 Odor. The odor shall be determined by evaporating to dryness 200 ml of liquid in a loosely covered 400 ml beaker with a fresh filter paper in the bottom. The cover is removed at the point of complete evaporation and the beaker is odor tested several times until it has warmed to above the freezing point of condensed water on the outside. Odors will be most prevalent when the beaker has warmed to nearly room temperature. Detection of any odors shall be cause for rejection of the lot.

4.6.3 Gaseous hydrocarbons. The gaseous hydrocarbons shall be determined in parts per million by volume as methane equivalent by one of the following methods. In cases of dispute the method in (c) shall be used as referee.

(a) By a flame ionization type analyzer. The range used shall be no greater than ten times the specified maximum gaseous hydrocarbon content expressed as methane. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals by the use of calibration gas standards.

(b) By a gas cell-equipped infrared analyzer capable of and operated so that its sensitivity for methane is either 0.5 ppm or 10 percent of the specified maximum amount whichever is greater. The

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equipment shall be operated in accordance with ASTM E168-67 or E334-68. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals with methane calibration gas standards at approximately 3.5 microns (3000 cm^{-1} wavenumbers) (the characteristic absorption wave length for C-H stretching).

(c) By oxidizing the hydrocarbons to carbon dioxide which may be determined by classical chemical methods or infrared equipment prepared as specific to carbon dioxide.

(d) By a gas chromatograph in accordance with 4.6.12.

4.6.4 Unsaturated hydrocarbons. The unsaturated hydrocarbons shall be determined in parts per million by one of the following methods. In cases of dispute the method in (b) shall be used as referee.

(a) For acetylene only, by a wet chemical (Illosvay-colorimetric) method in which the sensitivity for acetylene is at least twice the maximum amount specified.

(b) By a gas cell-equipped infrared analyzer capable of and operated so that its sensitivity for acetylene or ethylene as appropriate is at least twice the maximum amount specified. The equipment shall be operated in accordance with ASTM E168-67 or E334-68. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals with calibration gas standards at approximately 10.6 microns (950 cm^{-1} wavenumbers) and 13.7 microns (730 cm^{-1} wavenumbers) as appropriate.

(c) By a gas chromatograph in accordance with 4.6.12.

4.6.5 Halogenated hydrocarbons. The halogenated hydrocarbons shall be determined in parts per million by one of the following methods. In cases of dispute the method in (a) shall be used as referee.

(a) By an electronic type halide detector capable of and operated so that its sensitivity for halogenated solvents is at least twice the maximum amount specified. The analyzer shall be calibrated (zeroed and spanned) just prior to analysis of each sample using gaseous dilution calibration standards. This procedure shall be carried out only in an atmosphere free from contamination with halogens or their compounds.

(b) By a gas cell-equipped infrared analyzer capable of and operated so that its sensitivity for halogenated solvents is at least twice the maximum amount specified. The equipment shall be operated in accordance with ASTM E168-67 or E334-68. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals with halogenated solvents.

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(c) By a gas chromatograph in accordance with 4.6.12.

4.6.6 Carbon monoxide. The carbon monoxide shall be determined in parts per million by volume by one of the following methods. In cases of dispute the method in (b) shall be used as referee.

(a) By a catalytic combustion analyzer with a limit of detectability at least twice the maximum amount specified. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals with calibration gas standards using air as the base gas.

(b) By a gas cell-equipped dispersive or non-dispersive infrared analyzer capable of and operated so that its sensitivity for carbon monoxide is either 0.5 ppm or 10 percent of the specified maximum amount whichever is greater. The equipment shall be operated in accordance with ASTM E168-67 or E334-68. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals with calibration gas standards at approximately 4.6 microns (2175 cm^{-1} wavenumbers). Care should be exercised not to confuse the 4.5 micron peak for nitrous oxide with the 4.6 micron peak for carbon monoxide.

(c) By a gas chromatograph in accordance with 4.6.12.

4.6.7 Carbon dioxide. The carbon dioxide shall be determined in parts per million by volume by one of the following methods. In cases of dispute the method in (a) shall be used as referee.

(a) By a gas cell-equipped dispersive or non-dispersive infrared analyzer capable of and operated so that its sensitivity for carbon dioxide is 0.5 ppm or 10 percent of the specified maximum amount whichever is greater. The equipment shall be operated in accordance with ASTM E168-67 or E334-68. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals with calibration gas standards at approximately 4.3 microns (2325 cm^{-1} wavenumbers).

(b) By a gas chromatograph in accordance with 4.6.12.

4.6.8 Nitrogen dioxide. The nitrogen dioxide shall be determined in parts per million by volume by one of the following methods. In cases of dispute the method in (a) shall be used as referee.

(a) By a gas cell-equipped infrared analyzer capable of and operated so that its sensitivity for nitrogen dioxide is at least twice the maximum amount specified. The equipment shall be operated in accordance with ASTM E168-67 or E334-68. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals with calibration gas standards at approximately 6.2 microns (1613 cm^{-1} wavenumbers).

(b) By a gas chromatograph in accordance with 4.6.12.

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4.6.9 Sulfur dioxide. The sulfur dioxide shall be determined in parts per million by volume by one of the following methods. In cases of dispute the method in (a) shall be used as referee.

(a) By a gas cell-equipped infrared analyzer capable of and operated so that its sensitivity for sulfur dioxide is at least twice the maximum amount specified. The equipment shall be operated in accordance with ASTM E168-67 or E334-68. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals with calibration gas standards at approximately 7.3 microns (1370 cm^{-1} wavenumbers).

(b) By a gas chromatograph in accordance with 4.6.12.

4.6.10 Moisture. The moisture shall be determined as dew point in degrees fahrenheit or degrees centigrade, or parts per million by volume by one of the following methods. In cases of dispute the method in (a) shall be used as referee.

(a) By an electrolytic hygrometer on a range which is no greater than ten times the specified maximum moisture content.

(b) By a piezoelectric sorption hygrometer on a range which is no greater than ten times the specified maximum moisture content.

(c) By a frost point analyzer in which the temperature of a viewed surface, at the time frost first begins to form, is measured.

(d) By an analyzer built and operated in accordance with NBS Research Paper No. 1865.

4.6.11 Particulate. The particulate shall be determined in milligrams (mg) by passing five liters liquid air sample through a 1.2 micron-rated tared analytical filter disc contained in a suitable holder. The filtered liquid is collected and measured in an open container (dewar). The assembly is warmed and the disc reweighed.

4.6.12 Gas chromatographic method. The analyses shall be determined by the gas chromatographic method in accordance with ASTM E260-65T or D2504-67 as applicable, or as follows:

(a) The analyzer shall be capable of separating and determining the specified component with a sensitivity of 0.1 ppm or 20 percent of the specified maximum amount of the component, whichever is greater. Appropriate impurity concentrating techniques may be used to attain this sensitivity. The analyzer shall be calibrated (zeroed and spanned) at appropriate intervals by the use of calibration gas standards which contain the applicable limiting characteristic gaseous components in air.

5. PREPARATION FOR DELIVERY

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5.1 Packaging. The product furnished under this specification is a hazardous material as defined and regulated by the Department of Transportation (DOT) regulations. All packaging to be shipped commercially by any mode of transportation shall comply with the requirements of DOT regulations 49 CFR 171-190, or DOT special permit obtained in accordance with 49 CFR 170.13 by the shipper in conjunction with the Commander, Headquarters Military Traffic Management and Terminal Service Attn: Safety Division (TES), Washington, D.C. 20315. All packaging to be shipped by military air shall comply with DSAM 4145.3 (AFM 71-4). The product shall be packaged in containers and unit quantities as specified by the procuring activity (6.2 and 6.3).

5.2 Preparation of containers. Prior to filling, the contractor shall establish the condition of all containers to insure that they are:

(a) Free from contamination.

(b) Possess suitable vacuum in annular area if applicable.

(c) Have a current hydrostatic test date as required.

(d) Otherwise suitable for shipment and storage.

Contractor owned containers shall be cleaned, tested, and repaired by the contractor at his own expense. Leased or government owned containers shall be cleaned, tested, and repaired in accordance with the schedule established in the contract or purchase order.

5.2.1 Cleaning, testing, and repair. Unless otherwise provided for in the contract or purchase order, all containers shall be visually inspected internally and externally for the presence of water, rust, scale, oil film, or other foreign matter, and physical damage. Any physical damage which would endanger safe transportation of the product shall be repaired prior to reuse. If evidence of internal contamination is found, the containers shall be recleaned by a suitable method to remove the contamination prior to use. Hydrostatic testing shall be accomplished in accordance with DOT requirements. Tested containers shall be cleaned and dried prior to use.

5.2.2 Gaskets. Gaskets used to seal container openings shall be polytetrafluoroethylene, polychlorotrifluoroethylene, or other material compatible with the product and approved for use by the procuring activity. The contractor shall assure that all gaskets are serviceable and furnish new gaskets when necessary so that a tight seal is assured.

5.3 Filling. Containers shall be filled through a filter (3.4.2). Containers shall not be entirely filled. Sufficient space shall be left in each container to assure that no leakage or distortion of the container occurs as specified by DOT requirements.

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5.4 Labeling and marking. Each container shall be labeled and placarded in accordance with MIL-STD-129, and established DOT requirements or DOT special permit. In addition, an identification tag, and container color code shall be used.

5.4.1 Identification tag. Unless otherwise specified in the contract or purchase order, an identification tag impervious to climatic conditions shall be wired to the outlet port of each container and shall contain the following information: Product name, specification number with revision letter, grade designation, FSN number, quantity, name of manufacturer, name of contractor (if different from manufacturer), date of manufacture, and lot identification number.

5.4.2 Cylinder color code. Unless otherwise specified by the procuring activity, each cylinder shall be color coded in accordance with MIL-STD-172 and the exact name identification to be marked on the outside of the cylinder shall be "Liquid Air". Any other name identification shall be obliterated by removing or overpainting.

6. NOTES

This section contains nonmandatory provisions only to assist both the contractor and buyer in the proper understanding and utilization of this specification.

6.1 Intended use. The liquid air covered by this specification is intended for the procurement of two grades of liquid air for human breathing purposes as further described in 6.1.2 and 6.1.3. Physiological pressure limitations for end use are not covered by this specification and corrections must be made for deviations from sea level pressures when encountered.

6.1.1 Nature of liquid air. Liquid air is an unstable mixture of elements and compounds. As a consequence it is recognized that the values for the constituents are in constant change. The limiting values given in this specification are for the purpose of one time acceptance for procurement only. Changes in values subsequent to the procurement acceptance have to be controlled by use limits (6.4.2) established by the procuring activity. Generally used and accepted use limits for oxygen content by volume are a minimum of 18.25 percent and a maximum of 30 percent. In addition selected minor constituents may also be controlled for use limit purposes.

6.1.2 High purity. Grade A liquid air is intended for use in what could be considered as critical applications such as manned or unmanned testing of flight hardware, non-flight hardware, hardware which interfaces with flight hardware and prototype hardware, all of which require contamination controls established in accordance with other specifications. It may also be required where any special considerations dictate use of a high purity liquid air.

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6.1.3 Ordinary purity. Grade B liquid air is intended for use in ground support applications such as self-contained air breathing safety suit operations (Self-Contained Atmospheric Protective Ensamble (SCAPE) suits), toxic materials testing operations, or entry into tanks or any other environment where there may be contaminated or insufficient air supply where the equipment is not subject to contamination control requirements. It may also be used in such operations as air-sea rescue work, fire fighting, decompression chambers, or hyperbaric chambers when use conditions are duly analyzed and determined to be compatible for this grade of liquid air. For procurement purposes of this grade, minor constituents need not be controlled for physiological reasons because of the inherent self-limiting characteristics in the production of the liquid air. Their analysis as a procurement requirement would only serve to materially increase cost.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, grade, and date of this specification.

(b) Method of shipment, type, and capacity of containers (5.1 and 6.3).

(c) Quantity by weight in pounds (avoirdupois).

(d) When inspection requirements are to be performed by other than the supplier (4.1).

(e) When testing of samples is required at other than the manufacturing and filling location (4.3.2).

(f) When sampling method is to be other than that specified (4.3.2).

(g) When samples are to be taken from other than the shipping container as specified (4.3.2).

(h) When chemicals for tests are to be other than as specified (4.5.1).

(i) When analytical equipment is to be operated other than as specified (4.5.4).

(j) When cleaning, testing, or repair of containers is to be other than as specified (5.2.1).

(k) When the identification tag is to be other than as specified or waived (5.4.1).

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(1) When the container color code is to be waived (5.4.2).

6.2.2 Contract data requirements. Data conforming to Data Item Description DI-T-138-1, Quality Conformance Test Reports (Fuels), is a requirement for delivery in connection with this specification. The data item will be specified for delivery on the DD Form 1423. In addition to the copies required by the procuring activity, one copy of each report should be addressed to AFRPL (RPORS), Edwards, CA 93523.

6.3 Containers. As of the date of this specification, the DOT regulations do not authorize any specification containers for liquid air. All packaging requires DOT special permit. Requesting use of specification DOT 4L200 cylinders is suggested for unit quantities of 1000 pounds of water equivalent or less. Specially designed cryogenic transport and storage containers for military use may also be suggested for use. Samplers conforming to MIL-S-27626 or Cosmodyne Corporation, Model CS 4.4 may be used for sampling.

6.4 Definitions.

6.4.1 Dew point. Dewpoint is expressed in degress fahrenheit (^{o}F) at one atmosphere (760mmHg). To convert dewpoint to other units, see table III.

6.4.2 Use limit. The use limit is established by using activities for operational/storage minimum acceptability requirements as differenciated from the procurement limit which is established in a specification for procurement acceptance requirements. The use limit may be identical to, or different from a procurement limit to allow for a permissable degree of additional degradation of the limit while in operational/storage use. A use limit cannot be used as a basis for procurement acceptance.

6.5 Alternate publications. The following commercial publications may be used in lieu of or in conjunction with 49 CFR 171-190, 46 CFR 146, and 14 CFR 103 as appropriate as the content is identical or in the case of airlines is supplementary.

(a) Agent T. C. George's Tariff No. 23, Hazardous Materials Regulations of the Department of Transportation. Available from Association of American Railroads, Bureau of Explosives, 63 Vesey Street, New York, N. Y. 10007.

(b) Agent Fruend's Tariff, Hazardous Materials Regulations of the Department of Transportation. Available from American Trucking Associations, Inc., 1616 P Street, N. W. Washington, D. C. 20036.

(c) Tariff 6D, Transportation of Restricted Articles by Air. Available from Airline Tariff Publishers, Inc., 1825 K. Street, N. W. Washington, D. C. 20006.

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Table III Moisture Conversion Data
(all referred to 70°F & 14.7 psig)

Dew Point °F	Dew Point °C	ppm (by volume)	mg/liter
0	-17.8	1180	0.88
-5	-20.5	895	0.67
-10	-23.3	690	0.51
-15	-26.1	525	0.39
-20	-28.9	400	0.30
-25	-31.6	305	0.23
-30	-34.4	225	0.17
-35	-37.2	165	0.12
-40	-40.0	120	0.012
-45	-42.8	87.0	0.065
-50	-45.6	63.0	0.047
-55	-48.3	45.0	0.034
-60	-51.1	32.0	0.024
-65	-53.9	23.0	0.017
-70	-56.7	16.2	0.012
-75	-59.4	11.4	0.0085
-80	-62.2	7.8	0.0058
-85	-65.0	5.3	0.0040
-90	-67.8	3.5	0.0026
-95	-70.5	2.3	0.0017
-100	-73.3	1.5	0.0011
-105	-76.1	0.93	0.00069
-110	-78.9	0.58	0.00043

Custodians:

Air Force - 12

Review Activities:

Air Force - 19, 68, 70

Civilian Agency Interest:

NAS

Preparing Activity:

Air Force - 12

Project No. 6830-F068

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

MIL-A-27420(USAF) AIR, LIQUID, FOR BREATHING PURPOSES.

ORGANIZATION (of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-0378

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AFRPL (RPORS)
EDWARDS, CALIFORNIA 93523

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DEPARTMENT OF THE AIR FORCE

UNITED STATES AIR FORCE
OFFICIAL BUSINESS

Director
Air Force Rocket Propulsion Laboratory (RPORS)
Edwards, California 93523

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